

CS & IT ENGINEERING



Operating System

Synchronization

Lecture – 05

By– Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Semaphore

Topic

Wait() & Signal()

Topic

Counting Semaphore

Topics to be Covered



Topic

Questions on Semaphore

Topic

Solution Without Busy Waiting



[Question]



#Q. Consider a scenario where 2 processes are available: P1 and P2. The processes must be executed in order that first P1 should execute completely then only P2 can start. Write a piece of code to control the sequence of execution with minimum number of semaphores?

P1
wait(s1)

====
signal(s2)

P2
wait(s2)

====

Binary Semaphores $\Rightarrow 2$

$s1 = 1$
 $s2 = 0$

[Question]



#Q. Consider a scenario where 2 processes are available: P1 and P2. The processes must be executed in order that first P1 should execute completely then only P2 can start. Then again in the sequence P1, P2 runs in strict alternation. Write a piece of code to control the sequence of execution with minimum number of semaphores?

```
P1  
while (True)  
{  
    wait(s1)  
    ===  
    signal(s2)  
}
```

```
P2  
while (True)  
{  
    wait(s2)  
    ===  
    signal(s1)  
}
```

Binary semaphore

$s1 = 1$
 $s2 = 0$

[Question]



#Q. Consider a scenario where 3 processes are available: P0, P1 and P2. The processes must be executed in order P2, P0 and P1 only. Write a piece of code to control the sequence of execution with minimum number of semaphores?

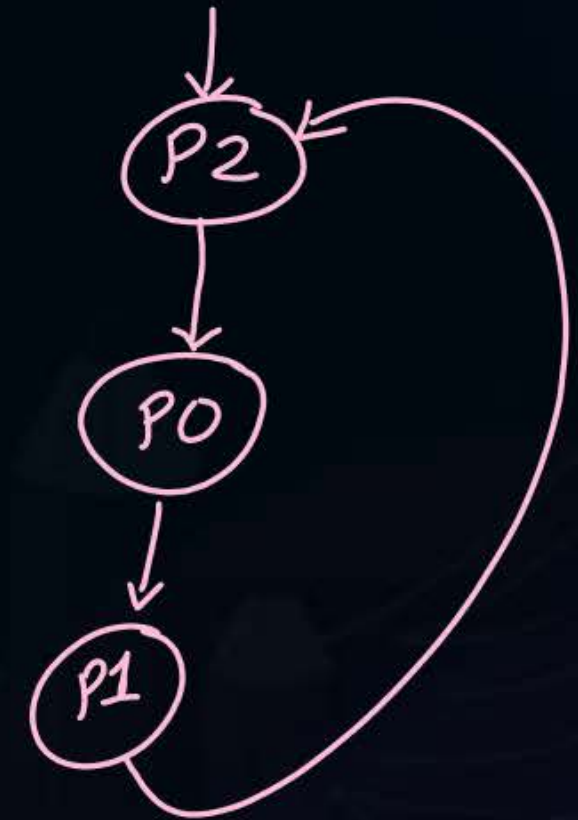
P0
{ wait(s0)
=
=
=
signal(s1)
}

P1
wait(s1)
=
=
=
signal(s2)

P2
wait(s2)
=
=
=
signal(s0)

3

$s_0 = 0$
 $s_1 = 0$
 $s_2 = 1$



[MCQ]



#Q. Consider the following threads, T_1 , T_2 , and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1 , S_2 , and S_3 , operated upon using standard `wait()` and `signal()`. The threads can be context switched in any order and at any time. Which initialization of the semaphores would print the sequence BCABCABCA...?

[2022]

T1	T2	T3
<pre>while(true) { wait (S₃); print("C"); signal(S₂); }</pre>	<pre>while(true){ wait(S₁); print("B"); signal(S₃);}</pre>	<pre>while(true) { wait(S₂); print("A"); signal(S₁); }</pre>

$T_2 \rightarrow T_1 \rightarrow T_3$

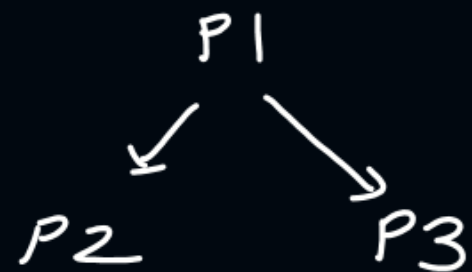
A $S_1 = 1; S_2 = 1; S_3 = 1$

B $S_1 = 1; S_2 = 1; S_3 = 0$

C ✓ $S_1 = 1; S_2 = 0; S_3 = 0$

D $S_1 = 0; S_2 = 1; S_3 = 1$

Ques)

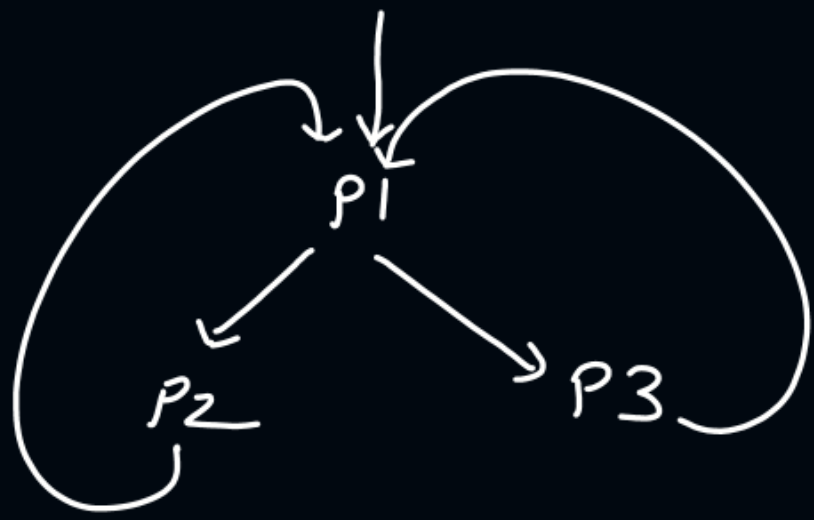


P2, P3 can start
only after P1 completes

use only binary semaphores

$S1 = 1, S2 = 0, S3 = 0$

P1	P2	P3
wait(S1)	wait(S2)	wait(S3)
≡	≡	≡
signal(S2)	≡	≡
signal(S3)	≡	≡



only binary semaphores

$$S_{11} = 1, S_{12} = 1$$

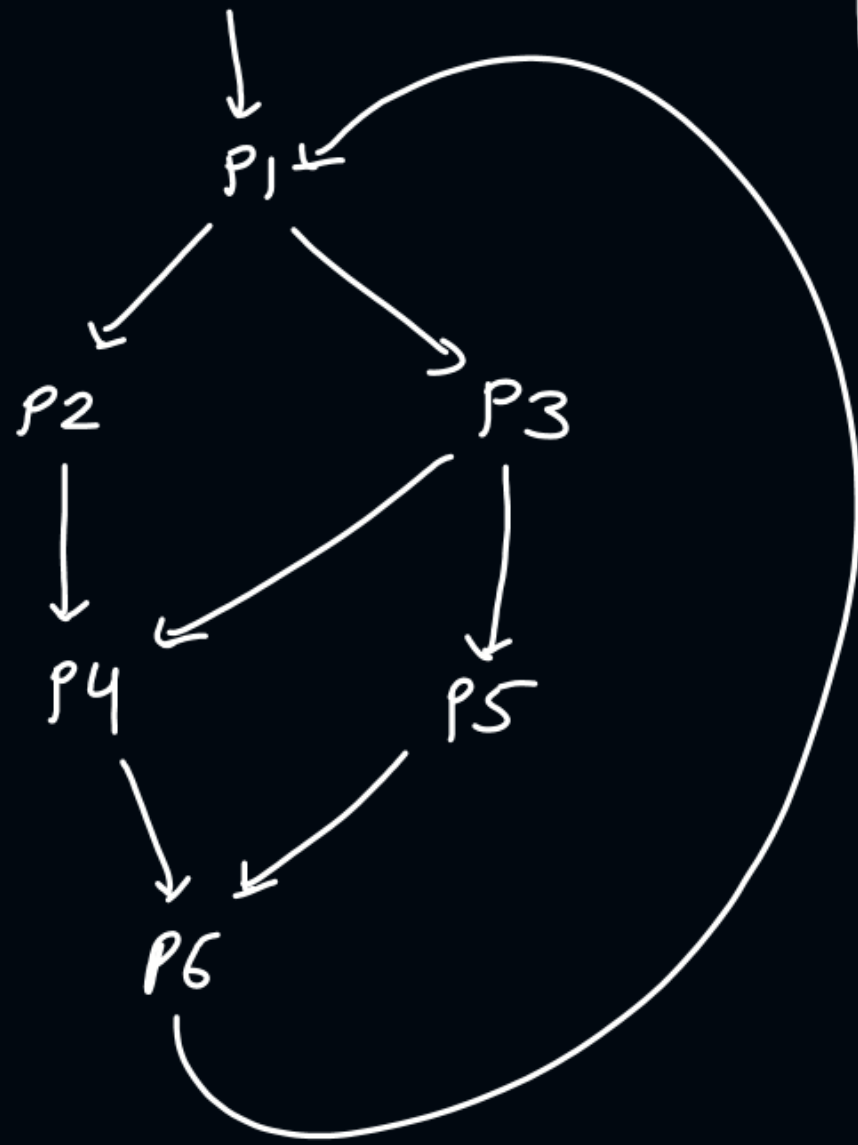
$$S_2 = 0$$

$$S_3 = 0$$

P1
 wait(S₁₁)
 wait(S₁₂)
 ≡
 signal(S₂)
 signal(S₃)

P2
 wait(S₂)
 ≡
 signal(S₁₁)

P3
 wait(S₃)
 ≡
 signal(S₁₂)



$S1 = 1, S2 = 0, S3 = 0, S41 = 0, S42 = 0, S5 = 0$
 $S61 = 0, S62 = 0$

P1	P2	P3	P4
wait(S1)	wait(S2)	wait(S3)	wait(S41)
=	=	=	wait(S42)
signal(S2)	=	=	=
signal(S3)	signal(S41)	signal(S42)	=
		signal(S5)	signal(S61)

P5	P6
wait(S5)	wait(S61)
=	wait(S62)
=	=
signal(S62)	signal(S1)

[NAT]



#Q. Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is 7?

[2016]

$$\text{Complete} = \underbrace{20}_{\text{Total}} - \underbrace{1}_{\text{blocked}} = 19 \quad \begin{matrix} P(S) \\ 12 \quad V(S) \end{matrix}$$

$$S + 12 - 19 = 0$$

$$\boxed{S = 7}$$

[NAT]



#Q. Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 43 $P(S)$ operations and 27 $V(S)$ operations are issued in some order. The largest initial value of S for which at least 5 $P(S)$ operation will remain blocked is 11?

successful $P(S) \Rightarrow 43 - S = 38$

$$S + 27 - 38 = 0$$

$$S = 11$$

$$S = 10$$

6 to be blocked

$$S = 9$$

7 — 11 —

$$S = 8$$

8 — 11 —

int $x = 2$

counting semaphore $S = 1$ $\Rightarrow$ mutual exclusion

$\Downarrow$

P_1, P_2 can run one-after-another

P_1
 $\text{wait}(S)$

$x = x + 2;$
 $\text{signal}(S)$

P_2
 $\text{wait}(S)$

$x = x - 3;$
 $\text{signal}(S)$

after P_1, P_2 completes

$\Downarrow$

$$x = 2 + 2 - 3 \\ = \underline{1}$$

int $x = 2$

Counting semaphore $S = 2 \Rightarrow$ no mutual exclusion

P1
wait(S)
 $x = x + 2$
signal(S)

P2
wait(S)
 $x = x - 3$
signal(S)

↓
Race condition
↓

$x + 2 - 3$

final value of $x = 4,$
-1,
1

[NAT]



Ans = 2

- #Q. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the process W and X reads x from memory, increments by one, stores it to memory and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory and then terminates. Each processes before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?
- [2013]

$$x = 0$$

$$s = 2$$

W

wait(s)

$x = x + 1$

signal(s)

X

wait(s)

$x = x + 1$

signal(s)

Y

wait(s)

$x = x - 2$

signal(s)

Z

wait(s)

$x = x - 2$

signal(s)

$$x + 1 + 1 - 2 - 2$$

$$x_{\max} = 0 + 1 + 1 = 2$$

$$x_{\min} = -4$$

no. of possible values of $x \Rightarrow 1, -2, 2, -1, -4, 0, -3$
distinct
7 values

Ans = 8

#Q. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the process W and X reads x from memory, increments by 2, stores it to memory and then terminates. Each of the processes Y and Z reads x from memory, decrements by 3, stores it to memory and then terminates. Each processes before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What are the total distinct possible values of x after all processes complete execution?

$$x + 2 + 2 - 3 - 3$$

$$2, -3, 4, -6, -1, 1, -4, -2$$

[MCQ]



#Q. Consider the two functions incr and decr shown below.

```
incr() {  
    wait(s);  
    X = X+1;  
    signal(s);  
}  
  
decr() {  
    wait(s);  
    X = X-1;  
    signal(s);  
}
```

$10 + 1 + 1 + 1 + 1 + 1 + 1 - 1 - 1 - 1 - 1$

There are 5 threads each invoking incr once, and 3 threads each invoking decr once, on the same shared variable X. The initial value of X is 10.

Suppose there are two implementations of the semaphore s, as follows:

I1: s is a binary semaphore initialized to 1. $\Rightarrow$ mutual exclusion $\Rightarrow 10 + 5 - 3 = 12$

I2: s is a counting semaphore initialized to 2. $\Rightarrow$ Race condition $\Rightarrow 10 - 1 - 1 - 1 = 7$

Let V1, V2 be the values of X at the end of execution of all the threads with implementations I1, I2, respectively.

Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively?

[2023]



15, 7



7, 7



✓ 12, 7



12, 8



2 mins Summary



Topic

Questions on Semaphore

Topic

Solution Without Busy Waiting



Happy Learning

THANK - YOU